Table: Employees

EmpID	EmpName	DepartmentID	Salary	HireDate
1	John Carter	10	60000	2018-04-12
2	Emma Stone	20	75000	2016-11-01
3	Ryan King	10	50000	2019-07-19
4	Sarah Lopez	30	82000	2020-01-03
5	Adam White	20	72000	2017-05-30

Table: Departments

DepartmentID	DepartmentName
10	IT
20	HR
30	Finance

Table: Projects

ProjectID	EmpID	ProjectName	HoursWorked
101	1	Alpha	120
102	2	Beta	90
103	3	Alpha	110
104	4	Delta	150
105	2	Gamma	60

SQL Practice Questions with Subqueries

1. Select all employees working in a department whose name contains ‘a’ (use subquery + LIKE).

2. Find employees whose salary is higher than the average salary of all employees (subquery + aggregate).

3. List employees whose department has more than 1 employee (subquery + GROUP BY).

4. Show employee names who worked on projects with more than 100 total hours (subquery).

5. Retrieve employees who are NOT assigned to any project (NOT IN subquery).

6. Get departments that have no employees (NOT IN + subquery).

7. Show employees hired after the earliest hired employee in the IT department (subquery + MIN).

8. List employees whose salary is greater than the highest salary in HR (subquery + MAX).

9. Find the employee(s) with the second-highest salary (subquery + TOP logic).

10. Show departments where the total project hours exceed 200 (subquery + group & aggregate).

11. List employees whose names start with the same letter as the highest-paid employee (subquery + LIKE).

12. Retrieve employees who share the same salary as at least one other employee (subquery + GROUP BY HAVING COUNT).

13. List employees who have worked more project hours than the average hours for their department (correlated subquery).

14. List all employees whose salary is between the minimum and maximum salary in Finance (subquery).

15. Retrieve departments where the average salary is above the overall company average (subquery + aggregate).

16. List employees working on the same projects as employee ‘Emma Stone’ (subquery).

17. List employees whose name length is longer than the average employee name length (LEN + subquery).

18. Show employees whose salary is a multiple of 5,000 and greater than the average salary for their department (math + subquery).

19. Get the employee who worked the maximum hours on any project (subquery + MAX).

20. Show employees whose project count is above the average project count (subquery + aggregate).

21. List employees in departments with the shortest department name (subquery + LEN).

22. Show employees earning salaries in the top 20% (subquery + percentile approximation).

23. Show employees whose salary is NOT IN the top 3 salaries in the company (subquery + NOT IN).

24. Retrieve employees whose department's name begins with the same first letter as the employee's name (subquery + LIKE).

25. List employees who worked on projects where hours worked is below the average for that project (correlated subquery).

26. Show employees whose salary is greater than the salary of all employees in Finance (ALL + subquery).

27. Show employees who earn more than at least one employee in HR (ANY + subquery).

28. Get employees whose hire date is earlier than the average hire date of their department (subquery).

29. Retrieve departments where the number of employees is NOT IN (2,3) (subquery + NOT IN).

30. Show employees whose salary rank is in the top 3 (subquery + ranking logic).

31. Find employees whose salary is above the average salary of their department (correlated subquery).

32. List employees whose hire year matches the earliest hire year in the company (YEAR() + subquery).

33. Retrieve employees working in departments where the department name appears more than once (GROUP BY + subquery).

34. Show employees whose salary equals the salary of at least one employee working in Finance (IN + subquery).

35. Show employee(s) who work on the most number of projects (subquery + COUNT).

36. List employees whose project hours are greater than the average project hours for employee ID 2 (subquery).

37. Find employees hired before every HR employee (subquery + ALL).

38. List employees whose salary is NOT the minimum salary of their department (subquery).

39. Show employees who share the same FIRST NAME as some other employee (string + subquery).

40. Retrieve employees whose salary is within 10% of the highest salary (math + subquery).

41. Show employees who have never worked on project 'Alpha' (NOT IN + subquery).

42. Get the department where employees have the longest average employee name length (LEN + subquery).

43. List employees whose salary is above the 75th percentile approximation (subquery using TOP & ORDER BY).

44. Show employees who earn more than the difference between the highest and lowest salaries in any department (math + subquery).

45. Retrieve employees whose EmpID appears in the top 50% of project assignments (subquery + COUNT / 2 logic).